

Deadlock Part=2

1. Deadlock Avoidance (Stopping deadlock before it happens)

Simple idea:

The system must know in advance what resources a process will need in the future.

This helps the system decide whether it should approve the request or make the process wait.

1.Safe State

A safe state means the system can give resources in such a way that all processes can finish without deadlock.

A **safe sequence** is an order in which all processes can run safely.

2.Unsafe State

Unsafe $\neq$ deadlock.

Unsafe simply means there is a chance of deadlock in the future.

Banker Algorithm

This algorithm checks whether giving a resource will keep the system safe or not.

Real-life Example (Bank Loan)

Before giving a loan, a bank manager checks:

“If I give this loan, will I still have enough money to give the minimum required loan to all other customers?”

If yes → approve

If no → ask the customer to wait

The OS does the same with resources.

2. Banker Algorithm (Safe/Unsafe check)

Working:

A process requests resources.

The OS checks:

“If I give this resource now, will the system still remain safe?”

- If yes → give the resource
- If no → make the process wait

Simple Real-life Example

A hostel has 10 buckets.

Three students need buckets to take a bath.

If the warden sees that giving 3 more buckets to P1 will leave no buckets for P2 and P3 later,
→ P1 is asked to wait.

This is exactly how the Banker Algorithm works.

3. Deadlock Detection (Finding deadlock after it happens)

If the system does not avoid or prevent deadlocks, it must detect them after they occur.

a. Single Instance (Wait-for Graph)

If each resource has only one instance, the OS creates a **wait-for graph**.

If there is a **cycle** → deadlock is present.

Real-life Example

P1 is waiting for P2

P2 is waiting for P3

P3 is waiting for P1

This round waiting is a deadlock.

Real world:

- A wants B's notebook
 - B wants C's notebook
 - C wants A's notebook
- No one can start → deadlock.

b. Multiple Instances

When multiple instances exist, the Banker Algorithm can be used for detection as well.

4. Recovery From Deadlock (How to get out of deadlock)

Once deadlock happens, the system must break it.

a. Process Termination

i. Kill all processes involved

Quick but all work is lost.

ii. Kill one process at a time

Stop one process, check again.

Repeat until the deadlock is removed.

Example:

Two users are stuck because printer and scanner are locked.

The IT team cancels job of one user → deadlock ends.

b. Resource Preemption

Take resources away from one process and give them to another process until the deadlock breaks.

Example:

Student A has a bucket and is waiting for a mug.

Student B has a mug and is waiting for a bucket.

Warden takes the mug from B and gives it to A → deadlock breaks.